

WINDHOEK, Namibia

WMO: 681100

Lat: 22.567S

Lon: 17.100E

Elev: 5659

StdP: 11.93

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	39.2	42.6	-5.1	5.2	66.4	0.6	7.0	64.9	18.2	57.9	16.0	59.0	7.5	90	0.337

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	24.3	93.6	58.8	91.7	58.2	90.0	57.7	65.7	78.1	64.7	77.2	63.8	76.2	6.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.8	105.6	68.1	61.4	100.5	67.6	60.6	97.6	67.5	34.1	77.8	33.1	77.1	32.3	76.0	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.0	13.8	12.1	DB	34.5	97.6	3.8	3.4	31.7	100.1	29.5	102.0	27.4	103.9	24.6	106.3	(4)
(5)			WB	27.7	67.8	2.3	2.9	26.0	69.9	24.6	71.6	23.3	73.3	21.6	75.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	69.3	75.4	74.8	72.2	68.3	64.1	59.6	58.8	63.3	70.2	74.0	75.0	76.2		
	DBStd	7.63	4.23	4.37	4.61	4.02	4.60	3.87	4.48	5.02	5.37	4.98	4.89	4.42		
	HDD50	9	0	0	0	0	2	1	4	2	0	0	0	0		
	HDD65	555	0	1	2	16	66	164	193	86	18	5	3	1		
	CDD50	7050	787	693	687	550	439	290	278	415	607	744	749	811		
	CDD65	2119	322	274	224	116	38	3	2	34	175	283	302	346		
	CDH74	20759	2951	2423	1879	952	429	74	79	474	1849	2959	3134	3556		
	CDH80	7941	1258	1002	658	169	39	3	3	44	594	1213	1353	1605		
(14) Wind	WSAvg	5.5	5.1	4.8	4.9	5.0	5.3	5.5	5.8	6.1	6.0	6.2	5.8	5.2		
(15) Precipitation	PrecAvg	14.6	3.0	3.8	3.3	1.3	0.2	0.0	0.0	0.0	0.1	0.5	1.1	1.3		
	PrecMax	27.6	11.9	7.7	7.0	4.5	1.0	0.6	0.0	0.2	1.1	2.3	4.2	4.6		
	PrecMin	5.4	0.1	0.2	1.2	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	5.1	3.0	2.2	1.9	1.3	0.3	0.2	0.0	0.3	0.6	1.1	1.1	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	95.2	94.2	93.1	86.3	83.9	78.9	79.0	83.4	91.6	93.6	94.8	96.6		
		MCWB	59.6	59.6	58.7	58.9	55.2	51.3	50.4	51.9	56.8	56.7	58.3	63.4		
	2%	DB	92.3	91.4	89.5	84.2	80.6	75.8	75.6	80.9	88.0	91.2	91.8	93.3		
		MCWB	58.9	59.8	58.5	57.5	53.5	49.7	48.7	50.8	54.2	55.6	57.2	59.6		
	5%	DB	89.8	89.2	86.6	82.2	78.5	73.7	73.7	78.9	85.8	89.2	89.8	90.8		
		MCWB	59.0	60.1	58.6	57.0	53.0	49.4	48.3	50.0	53.2	55.0	56.7	58.5		
	10%	DB	87.2	86.3	84.0	79.7	76.5	71.7	71.5	76.8	83.7	86.7	87.8	88.2		
		MCWB	59.2	59.9	58.8	56.8	52.4	48.7	47.4	49.1	52.4	54.4	56.1	57.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.3	67.3	66.2	64.0	57.9	53.9	52.6	54.3	60.5	60.7	63.5	69.8		
		MCDB	78.1	78.0	77.8	76.4	74.9	73.1	75.9	78.8	77.3	81.3	78.3	85.0		
	2%	WB	65.0	65.6	64.7	62.1	56.3	52.0	50.6	52.7	57.1	59.0	61.7	65.4		
		MCDB	77.4	76.9	76.4	74.5	75.0	71.9	72.5	77.1	81.7	80.0	78.3	81.5		
	5%	WB	64.2	64.6	63.5	60.7	54.8	50.8	49.3	51.5	55.5	57.7	60.5	63.4		
		MCDB	76.8	76.5	74.8	74.9	74.6	70.8	70.9	75.7	80.5	79.5	77.5	77.8		
	10%	WB	63.2	63.6	62.4	59.3	53.4	49.5	48.0	50.0	54.0	56.6	59.3	62.0		
		MCDB	75.5	75.5	73.5	74.0	73.6	69.4	69.4	73.9	78.8	79.4	77.4	77.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	22.6	22.1	21.6	22.6	24.0	24.8	25.4	26.0	25.9	25.2	24.9	24.3		
		MCDBR	26.0	25.7	24.9	24.4	25.1	26.1	26.6	26.8	26.3	26.2	26.3	26.7		
		MCWBR	8.8	9.4	9.1	9.7	10.7	11.9	12.3	12.1	11.4	10.8	10.5	9.8		
	5% WB	MCDBR	20.3	19.8	20.0	21.3	23.4	24.1	24.9	24.1	23.6	23.2	23.0	21.8		
		MCWBR	7.0	7.8	8.1	9.1	10.4	11.3	11.6	11.5	10.4	9.0	8.9	8.4		
(40) Clear-Sky Solar Irradiance	taub		0.336	0.335	0.327	0.299	0.266	0.245	0.253	0.298	0.365	0.374	0.321	0.322		
	taud		2.508	2.532	2.530	2.574	2.610	2.608	2.556	2.396	2.211	2.239	2.473	2.512		
	Ebn at Noon		319	316	310	306	305	307	307	301	292	300	322	325		
	Edh at Noon		36	35	33	30	26	25	27	35	45	46	37	36		
(44) All-Sky Solar Radiation	RadAvg		2344	2190	1982	1778	1616	1488	1558	1805	2102	2296	2471	2513		
	RadStd		155	168	110	110	36	32	24	52	66	96	132	136		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page